Syed Mohammed Umar Farooq

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Education

University of California San Diego (UCSD), USA **2025 – 2027**
M.S in Intelligent Systems, Robotics & Control, Electrical & Computer Engineering *3.96/4 GPA*

Indian Institute of Technology (IIT), Madras, India **2021 – 2025**
B.Tech in Electrical Engineering, Minor in Computing *3.71/4 GPA*

Experience

Graduate Researcher | *Sim-to-Real RL for Dexterous Grasping in Clutter* **June 2026 – present**
Advanced Robotics and Controls Lab (ARC Lab), Prof. Micheal Yip *UC San Diego*

- Built an Isaac Lab RL pipeline for a multi-fingered hand to retrieve objects from clutter with sim-to-real transfer.
- Designed the Isaac Lab environment (MDP, observations, actions, rewards) and trained PPO policies for manipulation.
- Developed a curriculum strategy with clutter-aware rewards for collision-free, target-selective grasping.
- Adapted BODex/UltraDexGrasp for the Inspire hand to generate grasp poses used as goal targets for reward shaping.

Graduate Researcher | *Gaussian Splatting Object SLAM* **Feb 2026 – June 2026**
Existential Robotics Lab (ERL), Prof. Nikolay Atanasov *UC San Diego*

- Developed an online object-level SLAM system on KITTI that represents scene objects as 3D Gaussian Splats, combining real-time object reconstruction with factor-graph-based localization.
- Integrated 3D Gaussian Splat object models into a Ceres factor-graph SLAM backend, using differentiable rendering to constrain camera and object poses from photometric consistency rather than geometric feature matches.
- Designed a retroactive factor insertion framework that associates newly reconstructed object models with historical keyframes and injects corresponding constraints into the factor graph, enabling global trajectory refinement without explicit loop-closure detection.

Projects

Visual-Inertial SLAM via Extended Kalman Filter | *Course Project, UCSD*

- Implemented a visual-inertial SLAM pipeline fusing stereo camera and IMU measurements using an Extended Kalman Filter on SE(3) for robot state estimation.
- Developed IMU propagation and stereo landmark update modules for joint trajectory estimation and 3D landmark mapping.
- Built full EKF-based VI-SLAM with joint pose-landmark optimization and evaluated trajectory correction against dead-reckoning drift.

LiDAR-Based SLAM with Pose Graph Optimization | *Course Project, UCSD*

- Developed a full SLAM pipeline combining encoder, IMU, LiDAR, and RGB-D sensing for localization, mapping, and texture reconstruction.
- Applied ICP-based 2D/3D scan alignment and occupancy grid mapping with Bresenham ray tracing to refine trajectory estimates and map structure.
- Integrated loop closure constraints in GTSAM factor graphs, improving global consistency of trajectories and occupancy maps.

LeRobot ACT Policy with Digital Image Processing | *Course Project, UCSD*

- Implemented robot imitation learning using the ACT policy in the **LeRobot** framework for teleoperated manipulation.
- Applied digital image processing techniques (CLAHE, gamma correction, shadow removal, deblurring) to improve perception under low-light, blur, and occlusion.
- Developed real-time scripts for policy execution, dataset recording, and camera processing to evaluate performance under visual degradation.

Q-Learning with Function Approximation | *B.Tech Project, IIT Madras*

- Implemented Q-learning and linear function approximation using 66-dimensional polynomial features to estimate Q-values in a large state-action space.
- Compared constant and Robbins-Monro learning rates, achieving better convergence with Robbins-Monro.
- Simulated a custom environment with stochastic transitions and a reward model $R(s, a) = s \cdot a$, benchmarking against dynamic programming results.

Technical Skills

Areas: Robotics, Reinforcement Learning, SLAM, Perception, Sensor Fusion & State Estimation, Motion Planning, Dexterous Manipulation & Grasping, Computer Vision, Deep Learning, Imitation Learning, Multi-threading
Programming Languages: C++, Python, JavaScript, MATLAB
Tools & Frameworks: ROS, ROS2, NVIDIA Isaac Sim, Isaac Lab, PyTorch, GTSAM, Ceres Solver, OpenCV, NumPy